Anatomy MCQ - Full Answer Script (CMU 1st Prof, 2021-2025)

Format: MFT questions (1-10, sometimes all 20) show the TRUE options and FALSE options by letter. SBA questions show the single best answer.

Method: answers follow standard textbook content (Snell, Gray's Anatomy for Students, Moore, BD Chaurasia, Langman, Junqueira, Ross). I checked the neuropore closure timing online (Langman-based sources: anterior ~day 25, posterior ~day 27-28). The rest is from textbook knowledge, not re-opened line by line. Red notes mark where I disagree with the Orbit key or the question is ambiguous - confirm those with your textbook or teacher.

The scan is a photo, so a few Orbit key marks may be misread by me. Where a note says Orbit differs, please compare with your page.

May 2025 - Paper I (Orbit p.76)

1	Fibrous joints	**T:** a, c | **F:** b, d, e
2	Collagen fibers abundant in	**T:** a, d | **F:** b, c, e
3	Spermatogenesis Note: Option e: reduction division (meiosis) is one step inside it, not the name of the whole process.	**T:** c, d | **F:** a, b, e
4	Middle ear communicates with	**T:** c, d, e | **F:** a, b
5	Extrapyramidal tracts	**T:** b, c, e | **F:** a, d
6	Limbic system includes	**T:** a, b, c, d | **F:** e
7	Uniaxial joint	**T:** a, d | **F:** b, c, e
8	MCA supplies Note: Central (lenticulostriate) branches supply basal ganglia and much of the internal capsule. Orbit's key differs; textbooks vary slightly.	**T:** a, b, d, e | **F:** c
9	Central lymphoid organ	**T:** a, e | **F:** b, c, d
10	Pterion: meeting point of Note: Also includes greater wing of sphenoid.	**T:** a, b, c | **F:** d, e
11	Muscles of facial expression	**b)** Buccinator
12	Parasympathetic nucleus Note: Orbit marks b (cochlear nucleus) - wrong.	**d)** Edinger-Westphal nucleus
13	Endocrine gland	**b)** Has rich blood supply
14	Retina Note: As printed, a is false (pigment epithelium is the OUTERMOST layer). Orbit marks a, so the question probably meant 'outermost'.	**c)** Blind spot has no photoreceptors
15	Cerebellum	**e)** Connected with brainstem by peduncles
16	Mast cells	**a)** Participate in allergic reaction
17	Skeletal muscle fiber	**c)** Multinucleated fibers with striations
18	Mature neurons are	**a)** Permanent cell
19	Nerve passing through cavernous sinus	**a)** Abducent
20	Basal nuclei	**e)** Caudate nucleus

May 2025 - Paper II (Orbit p.77)

1	Female urinary bladder arteries Note: In females the vaginal artery replaces the inferior vesical artery.	**T:** a, c, d | **F:** b, e
2	Modifications of GI mucosa	**T:** a, c, d, e | **F:** b
3	Lateral branches of abdominal aorta supply	**T:** a, b, e | **F:** c, d
4	Contents of adductor canal Note: Femoral artery, vein, saphenous nerve (and nerve to vastus medialis).	**T:** a, b, d | **F:** c, e
5	Dorsiflexion of foot Note: Orbit marks EDL as F and flexor hallucis longus as T - wrong. FHL is a plantar flexor.	**T:** a, b, d | **F:** c, e
6	Posterior wall of axilla formed by Note: Orbit marks subscapularis as F - wrong. Posterior wall: subscapularis, teres major, latissimus dorsi.	**T:** b, d, e | **F:** a, c

7	Vena caval opening of diaphragm Note: Orbit marks c T - wrong; the opening is widened (not closed) in inspiration. It lies at T8.	**T:** a, d | **F:** b, c, e
8	Intercostal nerves Note: 11 pairs (so 'in number' option a is F).	**T:** b, d, e | **F:** a, c
9	Conducting system of heart	**T:** a, b, d | **F:** c, e
10	Muscle supplied by axillary nerve Note: Orbit marks teres major and coracobrachialis T - wrong (lower subscapular and musculocutaneous nerves).	**T:** a, c | **F:** b, d, e
11	Deep inguinal ring lies in layer	**d)** Fascia transversalis
12	Ligament preventing backward falling of trunk Note: Orbit marks ischiofemoral. Standard texts (BD Chaurasia): iliofemoral (Y ligament) prevents the trunk falling backward.	**a)** Iliofemoral ligament
13	Muscle connecting vertebral column with lower limb	**d)** Psoas major
14	Urethral fold fails to fuse	**c)** Hypospadias
15	Organ completely covered by peritoneum Note: Except at the hilum.	**d)** Spleen
16	Artery in superior mediastinum	**d)** Left subclavian artery
17	Phrenic nerve supplies Note: Visceral pleura and visceral pericardium are not supplied by it.	**a)** Diaphragm
18	Vertebrosternal ribs Note: Vertebrosternal ribs are 1-7; only option a lies fully within 1-7.	**a)** 3, 4, 5
19	Submucous coat absent in	**a)** Gall bladder
20	Multipennate muscle	**e)** Subscapularis

November 2025 - Paper I (Orbit p.78)

1	Pterion: meeting point of Note: Orbit marks zygomatic T and frontal F - wrong. Pterion = frontal, parietal, temporal, greater wing of sphenoid.	**T:** a, b, e | **F:** c, d
2	Simple columnar epithelium present in	**T:** b, c | **F:** a, d, e
3	Extrapyramidal tracts	**T:** b, c, e | **F:** a, d
4	Refractive media of eyeball Note: Cornea (aqueous, lens, vitreous are the others).	**T:** b | **F:** a, c, d, e
5	Mitosis	**T:** a, c | **F:** b, d, e
6	Derivatives of mesoderm	**T:** b, e | **F:** a, c, d
7	Content of carotid sheath Note: Common/internal carotid artery, IJV, vagus. External carotid is outside the sheath. Orbit's key differs.	**T:** a, d, e | **F:** b, c
8	Purely motor cranial nerve	**T:** c, d | **F:** a, b, e
9	Central lymphoid organ	**T:** a, e | **F:** b, c, d
10	Deep cerebellar nuclei	**T:** b, c | **F:** a, d, e
11	Hard palate formed by	**c)** Maxilla
12	Muscle that opens laryngeal inlet	**e)** Thyroepiglottic
13	Parasympathetic nucleus Note: Nucleus ambiguus is branchial motor; mesencephalic nucleus is sensory.	**e)** Spinal (sacral parasympathetic nucleus)
14	Effect of sympathetic system	**d)** Decrease peristalsis
15	Reticular fiber found in	**c)** Lymphoid organ
16	Epithelial tissue	**a)** Is avascular
17	Principal muscle of mastication	**d)** Masseter
18	Cell in lacuna of Haversian system	**e)** Osteocyte
19	Nerve through cavernous sinus	**a)** Abducent
20	Layer of scalp = 'safety valve' haematoma	**c)** Loose areolar tissue

November 2025 - Paper II (Orbit p.79)

1	Branches of axillary artery	**T:** a, c, d | **F:** b, e
2	Parietal pleura	**T:** a, c, e | **F:** b, d
3	Abduction of shoulder Note: Serratus anterior (with trapezius) rotates the scapula for abduction beyond 90 degrees. Orbit marks e F - debated.	**T:** c, d, e | **F:** a, b
4	Tributaries of coronary sinus	**T:** a, d | **F:** b, c, e
5	Retroperitoneal organs	**T:** c, d, e | **F:** a, b
6	Mediastinal surface of right lung related to Note: Right phrenic nerve lies on the mediastinal pleura but is not a groove/impression. Orbit also F.	**T:** a, d | **F:** b, c, e
7	Content of spermatic cord	**T:** a, c, d | **F:** b, e
8	Supplied by superior gluteal nerve	**T:** a, c, d | **F:** b, e
9	Submucosal glands present in Note: Oesophageal glands proper lie in the submucosa (Junqueira). Orbit marks a F.	**T:** a, b | **F:** c, d, e
10	Derivatives of hindgut	**T:** a, e | **F:** b, c, d
11	Winging of scapula caused by paralysis of	**e)** Serratus anterior
12	Remnant of umbilical vein	**c)** Ligamentum teres hepatis
13	Branch arising from roots of brachial plexus	**d)** Long thoracic nerve
14	Diameter increased by pump-handle movement	**b)** Anteroposterior
15	Strongest ligament of hip joint	**c)** Iliofemoral ligament
16	Conducting system supplied by left coronary artery Note: Weak question: SA node is from RCA in about 60%, left coronary in about 40%. b is the best option here.	**b)** SA node
17	Viscera related to right kidney	**d)** Liver
18	Muscle forming inguinal ligament	**b)** External oblique
19	Primary support of uterus	**a)** Pelvic diaphragm
20	Keystone of medial longitudinal arch	**d)** Talus

May 2024 - Paper I (Orbit p.80)

1	Carotid sheath contains	**T:** a, c, e | **F:** b, d
2	Reticular fiber found in	**T:** a, b | **F:** c, d, e
3	Skeletal muscle fiber Note: Orbit marks e F; limb and trunk skeletal muscle arises from somites (paraxial mesoderm).	**T:** b, d, e | **F:** a, c
4	Structural abnormalities of chromosome Note: Ring chromosome is structural; Orbit marks it F. Turner and trisomy are numerical.	**T:** b, c, d | **F:** a, e
5	Fibrous joints	**T:** a, c | **F:** b, d, e
6	Commissural fibers	**T:** b, e | **F:** a, c, d
7	Nasopharynx communicates with Note: Orbit marks b F, but it communicates with the middle ear via the auditory tube.	**T:** a, b, c | **F:** d, e
8	Extrapyramidal tracts	**T:** b, c, e | **F:** a, d
9	Endodermal derivatives	**T:** a, b, c | **F:** d, e
10	Nasal septum formed by	**T:** a, c, e | **F:** b, d
11	Avascular structure in eyeball	**b)** Lens
12	Artery wall thicker than vein due to	**e)** Well developed tunica media
13	Descending tract	**d)** Tectospinal
14	Effect of sympathetic system	**d)** Decrease peristalsis
15	Nissl granules present in	**e)** Pyramidal cell
16	Parasympathetic fibers present in	**b)** Oculomotor nerve
17	Hard palate formed by	**c)** Maxilla
18	Cell in lacuna of Haversian system	**e)** Osteocyte

19	Multipolar neuron	**b)** Purkinje cell
20	Derivative of pharyngeal pouch	**b)** Thymus

May 2024 - Paper II (Orbit p.81)

1	Structure crossed by root of mesentery Note: Root crosses 3rd part of duodenum, aorta, IVC, right ureter, right psoas major. Orbit marks b F.	**T:** b, d | **F:** a, c, e
2	Veins ending into coronary sinus	**T:** a, b, d | **F:** c, e
3	Respiratory zone of bronchial tree Note: Begins at respiratory bronchiole; alveolar capillaries come from the pulmonary artery. Weakly worded question.	**T:** b, d | **F:** a, c, e
4	Flexor of elbow joint	**T:** a, d | **F:** b, c, e
5	Nerve related to humerus Note: Ulnar nerve touches the back of the medial epicondyle. Orbit's key differs (marks e T).	**T:** b, c, d | **F:** a, e
6	Quadriceps femoris	**T:** d, e | **F:** a, b, c
7	Histological features of duodenum	**T:** b, e | **F:** a, c, d
8	Contents of rectus sheath	**T:** a, b, e | **F:** c, d
9	Ureter Note: Ureter develops from the ureteric bud (mesoderm), not endoderm; it has three constrictions; lined by transitional epithelium. Orbit marks c T.	**T:** a, e | **F:** b, c, d
10	Femoral artery	**T:** b, c | **F:** a, d, e
11	Groove behind hilum of right lung lodges	**d)** Esophagus
12	Most common position of vermiform appendix	**b)** Retro-caecal
13	Most common site of clavicle fracture	**d)** Junction of medial two-thirds and lateral one-third
14	Transpyloric plane passes through	**d)** Origin of superior mesenteric artery
15	Hepatic lobule	**c)** Portal triad lies at corners of lobule
16	Lymphatics from nipple and areola drain mainly into	**a)** Pectoral group of axillary nodes
17	Back of medial epicondyle related to	**c)** Ulnar nerve
18	Paralysis of which nerve causes foot drop Note: Common peroneal nerve injury is the classic cause; of the options, deep peroneal.	**d)** Deep peroneal
19	Pleural reflection at mid-axillary line Note: Pleura: 8th rib (midclavicular), 10th (midaxillary), 12th (paravertebral).	**d)** 10th rib
20	Anterior relation of sacroiliac joint	**c)** Obturator nerve

November 2024 - Paper I (Orbit p.82)

1	Paired dural venous sinuses	**T:** a, e | **F:** b, c, d
2	Avascular structures Note: Orbit marks dermis T - wrong, the dermis is vascular.	**T:** b, d, e | **F:** a, c
3	Hyaline cartilage present in	**T:** a, c | **F:** b, d, e
4	Peripheral lymphoid organ Note: The spleen is a peripheral lymphoid organ; Orbit marks it F.	**T:** b, c, e | **F:** a, d
5	Hard palate formed by	**T:** a, b | **F:** c, d, e
6	Hind brain includes Note: Pons belongs to the hindbrain; Orbit marks it F.	**T:** a, c, e | **F:** b, d
7	Superior orbital fissure transmits	**T:** a, d, e | **F:** b, c
8	Spinal cord	**T:** a, b, e | **F:** c, d
9	Meiosis cell division Note: Meiosis occurs in primary spermatocytes/oocytes (immature germ cells); mature gametes do not divide. Orbit marks a T, b F - I disagree.	**T:** b, c, e | **F:** a, d
10	Deep cerebellar nucleus	**T:** b, c | **F:** a, d, e
11	Safety muscle of tongue	**b)** Genioglossus
12	Muscle that opens laryngeal inlet	**e)** Thyroepiglottic

13	Parasympathetic nucleus	**a)** Dorsal nucleus (dorsal motor nucleus of vagus)
14	Effect of sympathetic system	**d)** Decrease peristalsis
15	Bipolar neuron	**c)** Olfactory cell
16	Basal nuclei	**d)** Amygdaloid nucleus
17	Principal muscle of mastication	**d)** Masseter
18	Cell in lacuna of Haversian system	**e)** Osteocyte
19	Primary cartilaginous joint	**d)** First sternocostal joint
20	Characteristics of epithelium tissue	**T:** b, c | **F:** a, d, e

November 2024 - Paper II (Orbit p.83)

1	Opening of right atrium	**T:** a, b, c | **F:** d, e
2	Carpal tunnel contains	**T:** a, b, d | **F:** c, e
3	Great saphenous vein Note: It begins on the MEDIAL side of the foot, has valves, and passes in FRONT of the medial malleolus (Orbit marks e T).	**T:** a, b | **F:** c, d, e
4	Superior vena cava Note: SVC is formed by the right and left BRACHIOCEPHALIC veins.	**T:** b, d, e | **F:** a, c
5	Alveoli	**T:** a, c, d | **F:** b, e
6	Muscles that dorsiflex the foot	**T:** a, c, d | **F:** b, e
7	Intracapsular structure of knee Note: Popliteus tendon is intracapsular but extrasynovial. Orbit's key varies between years for the synovial membrane.	**T:** a, b, c | **F:** d, e
8	Extra-hepatic biliary apparatus	**T:** a, c, d | **F:** b, e
9	Pancreas Note: Alpha cells secrete glucagon (beta cells secrete insulin). Pancreas is mostly retroperitoneal.	**T:** a, b, e | **F:** c, d
10	Anterior compartment of leg Note: Flexor digitorum longus is in the posterior compartment; Orbit marks c T.	**T:** a, b, d | **F:** c, e
11	Strongest ligament of upper limb	**b)** Coracoclavicular ligament
12	Strong supinator of forearm	**b)** Biceps brachii
13	Jejunum and ileum develop from	**b)** Midgut
14	Parietal pleura	**b)** Pain sensitive
15	NOT a content of femoral sheath	**d)** Femoral nerve
16	Transpyloric plane passes through Note: The plane lies at L1 (some texts say lower L1/upper L2); L1-L2 is the closest option.	**b)** L1-L2
17	Referred pain of cholecystitis felt in	**c)** Tip of right shoulder
18	Least dilatable part of urethra	**c)** Membranous
19	Failure of testis to descend	**b)** Cryptorchidism
20	Structure crossing floor of ovarian fossa	**c)** Obturator nerve

May 2023 - Paper I (old curriculum) (Orbit p.84)

1	Membranous organelles Note: Ribosome is non-membranous; Orbit marks e T.	**T:** c, d | **F:** a, b, e
2	Mitotic cell division occurs in	**T:** c, e | **F:** a, b, d
3	Simple squamous epithelium present in	**T:** a, b, c | **F:** d, e
4	Surface ectodermal derivatives	**T:** c, d | **F:** a, b, e
5	Elastic cartilage present in Note: Auditory tube cartilage: textbooks vary; epiglottis is certain.	**T:** a, b | **F:** c, d, e
6	Secondary cartilaginous joint	**T:** a, c, d | **F:** b, e
7	Avascular structures	**T:** b, c, e | **F:** a, d

8	Surface modifications of cell Note: Villi and papillae are tissue-level, not cell-level, modifications.	**T:** a, b, c | **F:** d, e
9	Peripheral lymphoid organ	**T:** b, c, d | **F:** a, e
10	Connective tissue cell	**T:** b, c, d | **F:** a, e
11	Jugular foramen transmits Note: Also accessory nerve.	**T:** b, c, e | **F:** a, d
12	Muscles of mastication	**T:** a, b, d | **F:** c, e
13	Motor cranial nerve	**T:** a, c | **F:** b, d, e
14	Derivatives of 2nd pharyngeal arch	**T:** a, c, d | **F:** b, e
15	Parotid gland Note: It is an exocrine gland and is enclosed by parotid fascia; the duct opens in the vestibule opposite the upper 2nd molar. Orbit marks b F, c T, d T.	**T:** a, b, e | **F:** c, d
16	Hind brain formed by	**T:** a, c | **F:** b, d, e
17	Basal ganglia includes	**T:** a, c, e | **F:** b, d
18	Deep cerebellar nuclei	**T:** a, b, c | **F:** d, e
19	Occipital lobe of cerebrum Note: It lies above (not below) the tentorium cerebelli.	**T:** b, e | **F:** a, c, d
20	Refractive media of eyeball	**T:** a, c, e | **F:** b, d

November 2023 - Paper I (Orbit p.85)

1	Muscles of soft palate Note: Tensor tympani belongs to the middle ear. Orbit marks d T and e F.	**T:** b, e | **F:** a, c, d
2	Simple squamous epithelium present in Note: Peritoneum = mesothelium (simple squamous); Orbit marks it F.	**T:** a, b, d | **F:** c, e
3	Superior orbital fissure transmits Note: The optic nerve passes through the optic canal; Orbit marks d T.	**T:** a, b, c, e | **F:** d
4	Derivative of neural crest cell	**T:** b, c, e | **F:** a, d
5	Miniature long bone	**T:** c, d | **F:** a, b, e
6	Derivatives of pharyngeal pouches	**T:** b, c | **F:** a, d, e
7	Part of basal ganglia Note: Dentate nucleus is cerebellar; Orbit marks it T.	**T:** a, c, d | **F:** b, e
8	Sensory cranial nerve	**T:** d, e | **F:** a, b, c
9	Structures within parotid gland Note: Glossopharyngeal fibers reach the gland via lesser petrosal - otic ganglion - auriculotemporal nerve; the nerve itself is not within the gland.	**T:** a, b, d | **F:** c, e
10	Visual pathway formed by Note: Medial geniculate body is auditory (lateral geniculate is visual). Cornea is not part of the pathway proper; Orbit marks it T.	**T:** b, c | **F:** a, d, e
11	Extra-ocular muscle supplied by abducent nerve	**d)** Lateral rectus
12	Nucleus of paleocerebellum Note: Paleocerebellum (spinocerebellum) -> globose + emboliform; fastigial -> archicerebellum. Orbit marks c.	**e)** Globose and emboliform
13	Laryngeal cartilage made of both hyaline and elastic Note: Arytenoid: base hyaline, apex elastic. Epiglottis is elastic only.	**a)** Arytenoid
14	Betz cell present in	**d)** Internal pyramidal layer
15	Terminal branch of internal carotid artery Note: Middle cerebral is the other terminal branch.	**b)** Anterior cerebral artery
16	Pseudounipolar neuron located at Note: Trigeminal ganglion cells are also pseudounipolar, so b is arguable; c is the textbook example.	**c)** Dorsal root ganglion
17	Posterior neuropore closed at Note: Langman: anterior neuropore ~day 25, posterior ~day 27-28. Orbit marks 25th day.	**e)** 28th day
18	Safety muscle of larynx	**d)** Posterior cricoarytenoid
19	Unique feature of cervical vertebra	**b)** Foramen transversarium

20	DNA replication occurs in	**e)** S phase

November 2023 - Paper II (Orbit p.86)

1	Contents of middle mediastinum Note: Printed 'pulmonary nerve'; if 'pulmonary trunk' was meant, it is also T.	**T:** a, c | **F:** b, d, e
2	Carpal tunnel transmits Note: Palmaris longus passes superficial to the flexor retinaculum.	**T:** a, c | **F:** b, d, e
3	Anterior relations of right kidney	**T:** b, d, e | **F:** a, c
4	Muscles acting on both hip and knee	**T:** b, d | **F:** a, c, e
5	Left coronary artery (left posterior aortic sinus) Note: Orbit's key differs on c.	**T:** a, c, d | **F:** b, e
6	Muscles crossing shoulder joint	**T:** b, d, e | **F:** a, c
7	Pancreas Note: Alpha cells secrete glucagon, so d is F.	**T:** a, b, e | **F:** c, d
8	Parts of uterine tube	**T:** a, b, d, e | **F:** c
9	Structures crossed by root of mesentery	**T:** a, b, d | **F:** c, e
10	Opening of right atrium	**T:** a, b, c | **F:** d, e
11	Structure passing through venacaval opening	**b)** Inferior vena cava
12	NOT present in gastric gland	**b)** M cell
13	Retroperitoneal organ	**c)** Kidney
14	Continuation of axillary artery	**c)** Brachial artery
15	Loss of knee extension, injury to	**c)** Femoral nerve
16	Deltoid muscle	**a)** Is a multipennate muscle
17	Duodenum developed from	**c)** Foregut and midgut
18	Commonest type of vermiform appendix	**c)** Retrocaecal
19	Ilio-tibial tract is condensation of	**b)** Deep fascia of thigh
20	SA node located at	**d)** Crista terminalis

November 2022 - Paper I (Orbit p.87)

1	Duodenum (stem misprinted 'DNA found in') Note: The options describe the duodenum (compare Nov-2021 Paper II Q5). Orbit's key (only d T) does not match duodenum.	**T:** a, b, c, d | **F:** e
2	Gall bladder Note: Cystic artery arises from right hepatic artery. Orbit's key differs between papers.	**T:** a, b, c, e | **F:** d
3	Structures passing through sacral canal	**T:** a, b, d | **F:** c, e
4	Flexors of knee joint Note: Gracilis flexes and medially rotates the knee; Orbit marks it F.	**T:** b, c, d, e | **F:** a
5	Arch of aorta	**T:** a, b | **F:** c, d, e
6	Contents of spermatic cord Note: Cremasteric fascia is a covering, not a content.	**T:** a, b, c | **F:** d, e
7	Porto-systemic anastomoses	**T:** b, c | **F:** a, d, e
8	Parts of parietal pleura Note: Cervical pleura is a part (Orbit marks F). 'Peritoneal' is not pleura.	**T:** a, b, c, d | **F:** e
9	Structures crossed by root of mesentery	**T:** a, c, d | **F:** b, e
10	Branches of lumbar plexus Note: Nerve to pectineus comes from the femoral nerve (or accessory obturator); debatable. Pudendal is sacral plexus.	**T:** a, b, d | **F:** c, e
11	Nerves related to humerus	**T:** a, c, d | **F:** b, e
12	Structures opening in right atrium	**T:** a, c, d | **F:** b, e
13	Branches of posterior cord of brachial plexus	**T:** a, b, d, e | **F:** c
14	Structures related to ala of sacrum Note: Median sacral artery runs in the midline of the sacrum. Orbit's keys disagree between years.	**T:** a, b, c, d | **F:** e

15	Muscles supplied by musculocutaneous nerve	**T:** a, c, e | **F:** b, d
16	Umbilical cord contains	**T:** a, b, e | **F:** c, d
17	Intracapsular structures of knee	**T:** a, b, c | **F:** d, e
18	Derivatives of paramesonephric duct Note: Uterus and tubes; broad ligament is a peritoneal fold; lower vagina is partly urogenital sinus.	**T:** a, c | **F:** b, d, e
19	Structures passing behind inguinal ligament	**T:** a, b, c, d | **F:** e
20	Trachea Note: Bifurcates at T4 (sternal angle level).	**T:** a, b, c, d, e | **F:** -

November 2022 - Paper II (Orbit p.88)

1	DNA found in	**T:** b, d | **F:** a, c, e
2	Simple squamous epithelium present in	**T:** a, b, c | **F:** d, e
3	Non-membrane-bound organelles	**T:** a, c, e | **F:** b, d
4	Neural crest cell derivatives	**T:** a, b, c | **F:** d, e
5	Hyaline cartilage present in	**T:** d, e | **F:** a, b, c
6	Fibrous joint	**T:** a, b | **F:** c, d, e
7	Central lymphoid organ Note: Bone marrow and thymus are central; spleen is peripheral. Orbit marks bone marrow F and spleen T - wrong.	**T:** a, e | **F:** b, c, d
8	Osteoblast	**T:** a, e | **F:** b, c, d
9	Pneumatic bones Note: Frontal is pneumatic; Orbit marks it F.	**T:** b, c, e | **F:** a, d
10	Sperm	**T:** b, d, e | **F:** a, c
11	Carotid sheath contains	**T:** a, b, e | **F:** c, d
12	Sensory cranial nerve	**T:** c, d | **F:** a, b, e
13	Branch of external carotid artery	**T:** c, d | **F:** a, b, e
14	Derivatives of 1st pharyngeal arch	**T:** a, b, e | **F:** c, d
15	Muscles of soft palate	**T:** b, d | **F:** a, c, e
16	Brainstem includes	**T:** b, c, d | **F:** a, e
17	Parts of internal capsule	**T:** a, c | **F:** b, d, e
18	Deep cerebellar nuclei	**T:** b, c, d | **F:** a, e
19	Gray matter composed of	**T:** a, b | **F:** c, d, e
20	Retina contains cells	**T:** a, b, e | **F:** c, d

May 2022 - Paper I (Orbit p.89)

1	Mitotic cell division	**T:** b, d | **F:** a, c, e
2	Oogenesis	**T:** a, d | **F:** b, c, e
3	Turner syndrome	**T:** a, b | **F:** c, d, e
4	Results of fertilization	**T:** a, b, d | **F:** c, e
5	Characteristics of epithelial tissue Note: Epithelium regenerates well, is cell-rich, covers external AND lines internal surfaces, is avascular, and has a nerve supply. Orbit's key (b to e T) is wrong on c, d, e.	**T:** a, b | **F:** c, d, e
6	Synovial joint	**T:** a, c, e | **F:** b, d
7	Avascular structure	**T:** b, d, e | **F:** a, c
8	Morphologically short bones Note: Incus (ear ossicle) is not clearly a short bone; Orbit marks it T. Patella is sesamoid.	**T:** a, b, c | **F:** d, e
9	Morula	**T:** b, c, e | **F:** a, d
10	Peripheral lymphoid organ Note: Orbit marks bone marrow T and spleen/thymus F - wrong.	**T:** b, c, d | **F:** a, e
11	Taste sensation from tongue carried by	**T:** c, d | **F:** a, b, e

12	Carotid triangle contains	**T:** a, b, c, e | **F:** d
13	Mixed cranial nerve	**T:** a, b | **F:** c, d, e
14	Muscles of mastication	**T:** a, b, e | **F:** c, d
15	Paired dural venous sinuses	**T:** a, e | **F:** b, c, d
16	Hind brain includes	**T:** a, c, e | **F:** b, d
17	Cells in cerebral cortex	**T:** c, e | **F:** a, b, d
18	Sensory tracts of spinal cord	**T:** c, d, e | **F:** a, b
19	Commissural fibers	**T:** a, b | **F:** c, d, e
20	Muscle supplied by oculomotor nerve	**T:** a, b, d | **F:** c, e

May 2022 - Paper II (Orbit p.90)

1	Muscles forming rotator cuff	**T:** a, b, c, d | **F:** e
2	Structures related to neck of 1st rib	**T:** a, c, d | **F:** b, e
3	Fallopian tube	**T:** a, c, d, e | **F:** b
4	Muscles with double nerve supply	**T:** c, e | **F:** a, b, d
5	Structures related to transpyloric plane	**T:** b, d, e | **F:** a, c
6	Gluteus maximus	**T:** b, d | **F:** a, c, e
7	Fallot's tetralogy	**T:** a, b, d | **F:** c, e
8	Structures crossing pelvic brim Note: Ureter crosses at the bifurcation of the common iliac artery.	**T:** b, c, d | **F:** a, e
9	Portal triad contains Note: Triad = portal venule, hepatic arteriole, bile ductule. Bile canaliculi and central vein are not part of it. Orbit's key (c T, d F) looks swapped.	**T:** a, b, d | **F:** c, e
10	Muscles causing dorsiflexion of foot	**T:** a, b, c | **F:** d, e
11	The liver Note: Bare area is on the posterior part of the diaphragmatic surface, not 'superior'.	**T:** a, c, d, e | **F:** b
12	The pancreas	**T:** a, c, d, e | **F:** b
13	Structures related to base of heart	**T:** a, e | **F:** b, c, d
14	The kidney	**T:** a, c, d | **F:** b, e
15	Ureter	**T:** a, b, c, d | **F:** e
16	Aortic opening of diaphragm transmits	**T:** a, b, c | **F:** d, e
17	The spleen Note: Spleen lies deep to ribs 9-11.	**T:** a, c | **F:** b, d, e
18	Contents of femoral sheath	**T:** b, c, d, e | **F:** a
19	Foot drop is due to injury to Note: Common peroneal nerve.	**T:** c | **F:** a, b, d, e
20	Conducting system of heart	**T:** a, b, d | **F:** c, e

November 2021 - Paper I (Orbit p.91)

1	Phagocytic cells Note: Osteoclast (not osteocyte) is phagocytic; Orbit marks d T.	**T:** a, b, e | **F:** c, d
2	Meiotic type of cell division	**T:** a, b, d | **F:** c, e
3	Connective tissue cells	**T:** a, c | **F:** b, d, e
4	Neural crest derivatives	**T:** b, d, e | **F:** a, c
5	Ball and socket joint	**T:** c, e | **F:** a, b, d
6	Pneumatic bones Note: Temporal bone (mastoid air cells) is pneumatic; Orbit marks it F.	**T:** b, c, e | **F:** a, d
7	Central lymphoid organs	**T:** a, c | **F:** b, d, e
8	Fibrocartilage present in Note: IVD annulus fibrosus is fibrocartilage (Orbit marks c F). Inferior tibiofibular is a fibrous syndesmosis.	**T:** a, c, d | **F:** b, e

9	Collagen fibers abundant in	**T:** a, d | **F:** b, c, e
10	Multipennate muscles Note: Soleus and rectus femoris are bipennate. Orbit's key differs.	**T:** b, e | **F:** a, c, d
11	Motor cranial nerve	**T:** a, c | **F:** b, d, e
12	Derivatives of 1st pharyngeal arch	**T:** a, d, e | **F:** b, c
13	Branches of external carotid artery Note: Anterior auricular arteries arise from the superficial temporal (an ECA terminal branch) - indirect. Inferior thyroid is from thyrocervical trunk.	**T:** a, c, e | **F:** b, d
14	Parasympathetic fibers present in Note: Vagus also carries them.	**T:** b, c, d | **F:** a, e
15	Paired cartilage of larynx	**T:** c, e | **F:** a, b, d
16	Gray matter composed of	**T:** a, c, e | **F:** b, d
17	Brainstem includes	**T:** a, c, e | **F:** b, d
18	Deep cerebellar nuclei	**T:** a, c, d | **F:** b, e
19	Parts of internal capsule Note: Rostrum, trunk and splenium are parts of the corpus callosum. Internal capsule: anterior limb, genu, posterior limb, retrolentiform, sublenticular.	**T:** b, d | **F:** a, c, e
20	Refractory media of eyeball	**T:** b, d, e | **F:** a, c

November 2021 - Paper II (Orbit p.92)

1	Gall bladder	**T:** a, b, c, e | **F:** d
2	Intercostal nerves Note: Lower intercostals supply the peripheral diaphragm; fibrous pericardium is supplied by the phrenic nerve.	**T:** a, b, c, d | **F:** e
3	Lesser omentum contains Note: Free border: hepatic artery proper, bile duct, portal vein. IVC and coeliac trunk are not contents.	**T:** a, b, c | **F:** d, e
4	Structures opening in right atrium	**T:** a, c, d | **F:** b, e
5	Histological structures of duodenum Note: Peyer's patches are ileal.	**T:** a, b, c, d | **F:** e
6	Structure passing through sacral canal	**T:** a, b, d | **F:** c, e
7	Structures related to ala of sacrum Note: See note in Nov-2022 Paper I.	**T:** a, b, c, d | **F:** e
8	Branches of lumbar plexus	**T:** a, b, d | **F:** c, e
9	Contents of spermatic cord	**T:** a, b, c | **F:** d, e
10	Derivatives of paramesonephric duct Note: Round ligament of uterus comes from the gubernaculum; Orbit marks e T.	**T:** a, c | **F:** b, d, e
11	Nerves related to humerus	**T:** a, c, d | **F:** b, e
12	Common congenital anomalies of foot	**T:** a, c, d | **F:** b, e
13	Intracapsular structures of knee joint	**T:** a, b, c | **F:** d, e
14	Structures passing behind inguinal ligament	**T:** a, b, c, d | **F:** e
15	Radial nerve supplies Note: Anconeus is supplied by the radial nerve (Orbit marks b F).	**T:** b, c, d, e | **F:** a
16	The clavicle	**T:** a, c, e | **F:** b, d
17	Muscles innervated by femoral nerve	**T:** a, c, e | **F:** b, d
18	Venacaval opening of the diaphragm Note: At T8 (esophageal opening T10, aortic T12). Left phrenic pierces the muscle.	**T:** a, b | **F:** c, d, e
19	The female pelvis	**T:** b, c, e | **F:** a, d
20	Invertors of foot Note: EHL is mainly a dorsiflexor.	**T:** b, e | **F:** a, c, d

May 2021 - Paper I (Orbit p.93)

1	Membranous organelles	**T:** a, b, c | **F:** d, e
2	Mitotic cell division occurs in	**T:** b, c, e | **F:** a, d

3	Surface ectodermal derivatives	**T:** a, d | **F:** b, c, e
4	Primary cartilaginous joints	**T:** b, e | **F:** a, c, d
5	Paranasal air sinuses	**T:** a, b, d | **F:** c, e
6	Surface modifications of cell Note: Flagella are counted as a surface modification (as in May-2023); Orbit marks it F here.	**T:** a, b, e | **F:** c, d
7	Elastic cartilage present in	**T:** a, d | **F:** b, c, e
8	Peripheral lymphoid organs	**T:** a, b, e | **F:** c, d
9	Functional end artery present in	**T:** b, d, e | **F:** a, c
10	Osteoblast	**T:** c, d, e | **F:** a, b
11	Sensory cranial nerve	**T:** a, c | **F:** b, d, e
12	Contents of carotid sheath	**T:** b, c, e | **F:** a, d
13	Muscles of mastication	**T:** a, e | **F:** b, c, d
14	Oropharynx	**T:** a, c | **F:** b, d, e
15	Muscles of soft palate	**T:** c, e | **F:** a, b, d
16	Frontal lobe of cerebrum contains	**T:** b, c, d | **F:** a, e
17	Hind brain formed by	**T:** a, c | **F:** b, d, e
18	Basal ganglia includes	**T:** a, c, e | **F:** b, d
19	Effects of cerebellar lesion	**T:** b, d, e | **F:** a, c
20	Oculomotor nerve innervates	**T:** a, d, e | **F:** b, c

May 2021 - Paper II (Orbit p.94)

1	The liver Note: Bare area is on the posterior part of the diaphragmatic surface.	**T:** a, c, d, e | **F:** b
2	Structures crossed by root of mesentery	**T:** a, c | **F:** b, d, e
3	Hindgut derivatives	**T:** a | **F:** b, c, d, e
4	Structures opening in right atrium	**T:** a, c, d | **F:** b, e
5	Flexors of knee joint	**T:** b, c, d, e | **F:** a
6	Structures passing through sacral canal Note: Orbit marks obturator nerve T - wrong.	**T:** a, b, d | **F:** c, e
7	The kidney	**T:** a, c, d | **F:** b, e
8	Claw hand injury involves Note: Ulnar nerve. Printed 'thoracic duct' is unrelated (maybe 'thoracic outlet' misprint); Orbit marks c T.	**T:** b | **F:** a, c, d, e
9	The fallopian tube	**T:** a, c, d, e | **F:** b
10	Umbilical cord contains	**T:** a, b, e | **F:** c, d
11	Histological structures of duodenum	**T:** a, b, c, d | **F:** e
12	Arch of the aorta	**T:** a, b | **F:** c, d, e
13	Intracapsular structures of knee joint Note: Synovial membrane lines the capsule; Orbit's key varies between years.	**T:** a, b, c | **F:** d, e
14	Structures passing behind inguinal ligament	**T:** a, b, c, d | **F:** e
15	Tributaries of superior vena cava Note: Hemiazygos drains into the azygos vein, not directly into SVC. Small pericardial/mediastinal veins do drain directly.	**T:** a, d, e | **F:** b, c
16	Structures related to transpyloric plane	**T:** b, d, e | **F:** a, c
17	Muscles supplied by obturator nerve	**T:** a, d, e | **F:** b, c
18	Aortic opening of diaphragm transmits Note: Thoracic duct also passes through it.	**T:** a, b | **F:** c, d, e
19	The female pelvis	**T:** b, c, e | **F:** a, d
20	Contents of spermatic cord	**T:** a, b, c | **F:** d, e